

```

3- 4 2"B I B #@ MGC-PEA.REPAIRED-BETAVR.2.1
@ - *2# 5I- 4 2" 2 #'! +%1 2# 3 2 A%0 '2!@ 7H-!B" - D %L+%1 C B
# 1 2# #1 "L*4 A%0 #0'1 4-8 # L 3#8 - #0 Microgrid/#0 1 @ G
1. 2 #'!B #@
B #@ 5I@ G @'G A- A Static Frontend 5H@ 4 2 index.html A%0
B "@ 7H-! H- #4 2# Firebase 1H Client B " # D IA H Authentication
Storage
D %L*3 1 !5 1 5I
D %L + I2 5H
---
index.html B # *#I2 + I2@'G , Layout, Modal, Tab, Image map,
A%0B+% main.js
main.js Logic I! +! - #0 @ H Login, *4 4L 9IC I, CRUD I-!9
Import/Export, Dashboard, Report A%0 Realtime listener
#9 2 .jpg/.png/.bmp C I@ G A 1 / 2 #0 - - A H%0D LA%0B%
D %L .docx C I@ G A -#L!/#2" 2 I A *3+#1 2#A I -8 # L 3#8
README.md 7H-B #@ A "H-
2. * 2 1 " ##!B "#'!
9IC I@ 4 + I2@'G H2 @ #2'L@ -#L 2 1I index.html B+% CSS/JS
@!7H- 9IC I Login I'" Google #0 0-H2 *4 4L 2 Firestore collection
A%I'@ 4 @! 9 2! Role +%1 2 1I I-!9%-8 # L 0 9 -H2 /@ 5" C Fires
2"C IB # *#I2 sites/{siteKey}/devices/{deviceName} A%0D %LA 0
Firebase Storage
B # *#I2 I-!9%+%1 B "*#8 7-
'text
Firebase
+- users/{email}
| +- role: admin editor engineer viewer
| +- allowedSites: string[]
| +- profile fields
+- sites/{siteKey}
| +- devices/{deviceName}
| | +- records: #0'1 4 2# 3#8 / H-!/#1 #2
| | +- assetInfo: I-!9% #1 "L*4 A%0 #0 1
| +- registry/config
+- activityLogs/{logId}

```

```

py'
py
py3. index.html 3-0D#
py
py3.1 *H' +1'A%0 Style
py
pyD %L index.html 3+ 2)2D " (lang="th"), viewport, title, B+% Google
pySweetAlert2 CSS A%0 Tailwind CDN 2 1l 3+ tailwind.config @ 7H
pyA%0*5 - A # L/* 2 0
py
pyCSS 2"C D %LC l 3+ + l2 2 @ H Header A gradient, Card shadow
py 8H!+%1 , Badge * 2 0 #0 1 , Tab A%0B+! viewer-mode
py 5H H- 2 /A 1 *3+#1 9lC l 5HD!H!5*4 4L 9 l-!9%@ 4 2
py
py3.2 B # + l2@'G
py
py*H' Body A H @ G - L #0 - +%1 :
py
py1. Header *3+#1 Login/Logout, A* 7H- 9lC l, Role A%0 1' 1 Auto l
py2. loginPrompt *3+#1 A* l- '2! H- Login
py3. appContent *3+#1 @ 7l-+2+%1 +%1 Login
py4. 1'@%7- D L @ H @ 20 0%'", A!H*0@#5" , @ , #l2'
py5. Tab +%1 3 + l2: A 1 , #0'1 4 3#8 , #2" 2# #1 "L*4
py6. Modal +%2" 8 @ H 1 6 @+ 8 2# L, l-!9% #1 "L*4 , 1 2# 9lC l
pyReport, A%0 1 2# %8H! #1 "L*4
py
py3.3 A 1 -8 # L
py
py+ l2A 1 C l#9 2 A%0 HTML image map (<map>/<area>)
py@ 7H-C+l 9lC l %4 #4@' -8 # L 2 A%l'@ 4 -#L! 1 6 l-!9% - -8 #
py - 2 5l"1 !5 overlay-container-* *3+#1 '2 #- * 2 0 -8 # L @ H 4
py
py3.4 D% #2#5 2" -
py
py l2"D %L index.html B+% D% #2#5@+%H2 5l:
py
py- Firebase v8: app, firestore, auth, storage
py- SweetAlert2: popup/alert
py- Chart.js: #2 Dashboard
py- image-map-resizer: #1 scale image map @!7H-+ l2 -@ %5H"
py- PizZip + Docxtemplater + FileSaver: *#l2 / 2' LB+% #2" 2 Word
py- SheetJS/XLSX: Import/Export Excel
py- main.js: Logic - B #@
py
py4. main.js 3-0D#

```

```

๒
๒4.1 2#@ 7H-! H- Firebase
๒
๒ - I D %L 3+ firebaseConfig A%I' initialize Firebase I2"1 D!H@ "
๒ 2 1I *#I2 1'A #-I2 -4 #4 2#+%1 :
๒
๒- db *3+#1 Firestore
๒- auth *3+#1 Firebase Authentication
๒- storage *3+#1 Firebase Storage
๒- devicesCol *3+#1 collection devices @ 4!/*H' %2
๒
๒4.2 1'A #* 2 0 - A-
๒
๒#0 @ G state 1H Client @ H D L 1 8 1 (currentSiteKey), -8 #
๒(currentDevice), record 5H 3%1 A I (editIndex), + I2 1 8 1 - 2#2 ,
๒role, allowed sites, I-!9% profile A%0 cache * 2 0-8 # L
๒
๒4.3 1 L 1 '1 5HA%0@'%2
๒
๒!5 1 L 1 A % '1 5H+%2"#9 A @ 7H-#- #1 I-!9%D "A%0 I-!9% 2 Exc
๒
๒- formatThaiDate() A* '1 5H@ G .(.
๒- formatThaiDateTime() A* '1 @'%2A D "
๒- parseThaiDateToStandard() A % '1 5HD "/Excel/ISO @ G YYYY-MM
๒- parseThaiDateTimeToTS() A % '1 @'%2@ G timestamp
๒
๒4.4 *4 4L 2#C I 2
๒
๒#0 A" *4 4L I'" Role:
๒
๒- admin: 3D I 8 -"H2 #'! 6 1 2# 9IC IA%0 I-!9% #1 "L*4
๒- editor: A ID I-!9% 2!D L 5HD I#1 *4 4L
๒- engineer: #1 #2 / 4 2 H-! 2!D L 5HD I#1 *4 4L
๒- viewer: 9 I-!9% 3 1 A%0 9 H- I-!9%@ 4 2 2 *H'
๒
๒ 1 L 1 +%1 7- hasWriteAccess(), hasEngineerSiteAccess(), canAckn
๒canMarkFixed() 6H 9 @#5" H- @ 4 8H!/ 1 6 / 3@ I2/A ID I-!9%
๒
๒4.5 I-!9%D LA%0#2" 2#-8 # L
๒
๒ 1'A # sites @ G master data - D L 1I +! B "A H%0D L!5 name A%
๒@ 20 0%'", A!H*0@#5" , @ A%0 #I2' *H' OTHER_SUBDEVICES C I
๒other A%I' I- #0 8 sub-device @ 4H!@ 4! A%0 sitePrefixes C I*#I2 #-
๒KPL-000001
๒

```

py4.6 2#-H2 /@ 5" Firestore

py

py 1 L 1 %8H! 5l@ G Data Access Layer A H2":

py

py- getSiteCollection(siteKey) 7 H2 collection sites/{siteKey}/devices

py- getDeviceRecords(siteKey, device) -H2 records - -8 # L

py- saveDeviceRecords(siteKey, device, records) 1 6 records %1 F

py- getAllDevicesDocs(siteKey) -H2 @- *2#-8 # L 1l +! C D L

py- generateAutoId(siteKey) +2@% customId %H2*8 A%l'*#l2 @% 1 D

py

py+%1 2# 7-@- *2# 1 1'A -8 # L 1 #2" 2# A%0 field records @ G ar

py - #0'1 4 2# 3#8 / H-!

py

py4.7 Modal 1 6 @+ 8 2# L

py

py@!7H- 9lC l %4 -8 # L 2 A 1 +#7- 2#2 #0 @#5" openForm(device

pycurrentDevice, B+% l-!9% #1 "L*4 , B+% #0'1 4 A%0A* modal -#L

py 2"C -#L!!5 l-!9%*3 1 D lA H 9lA l , * 2 0 , '1 5H@ 4 @+ 8 , #2"%0(

py'1 5H H-!@*#G , H2C l H2", @% 5HC *1H , @- *2#-l2 -4 , '4 5A ID

py

py 2# 1 6 C l saveData() 6H #' *4 4L, #' l-!9% , -1 B+% D %L l2!5,

pyrecord C array, 1 6 Firestore, *#l2 log, -1 @ 2#2 /overlay A%0*

pynotification 2!@ 7H- D

py

py4.8 2# 1 2# l-!9% #1 "L*4

py

py*H' Asset modal C l@ G l-!9% #0 3-8 # L @ H Serial, Model, PEA

py* 2 5H 4 1l , #9 2 , '1 @#4H!/*4l *8 #0 1 #0 3 ' * 2 0 #0 1 l'"

pygetWarrantyStatus() A%0A* % l'" getWarrantyStatusHTML()

py

pyloadAssetData() B+% assetInfo 2 Firestore *H' saveAssetData() :

py %1 D C @- *2#-8 # L@ 4!

py

py4.9 2#2 *#8 A%0 Dashboard

py

pyupdateDeviceSummary() -H2 l-!9%-8 # L 1l +! - D L 1 8 1 *#8 3

py 4, !5 1 +2 A%l'*#l2 A 'C 2#2 2! filter/search/sort/date range - 2

pyrecord l'" 2# %4 A 'D l

py

pyrenderDashboardCharts() C l Chart.js *#l2 #2 @ H -1 1 -8 # L 5H

pyA%0@'%2@ %5H" H-!

py

py4.10 Overlay * 2 0 A 1

py

pyupdateDeviceStatusOverlays(siteKey, useCache) -H2 * 2 0%H2*8 - -

pyA%l'*#l2 element overlay l- 2 A 1 * 2 0+%1 7-:
py
py- normal: #- @ 5''
py- abnormal: #- *l! #0 #4
py- down: *1 %1) L 2 2 A #0 #4
py
pyCSS C index.html @ G 1' 3+ *5/animation - overlay *H' main.js
py@ G 1' 3 ' 3A+ H A%0* 2 0
py
py4.11 Realtime listener
py
pysetupRealtimeListener(siteKey) C l Firestore onSnapshot() @ 7H- 1
pycollection -8 # LC D L 1 8 1 @!7H- l-!9%@ %5H" 0@#5" updated
pyupdateDeviceStatusOverlays() 3C+l 2#2 A%0A 1 -1 @ 1 5
py
py4.12 Import/Export Excel
py
pyimportData(event) -H2 D %L .xlsx l'" SheetJS A%l'A" l-!9% 2 shee
py #0'1 4 2# 3#8 A%0 l-!9% #1 "L*4 2 1l *H C+l processAndSaveImpo
pybatch % Firestore A%0 l2! record 5H!5 ID l3
py
pyexportAllDataExcel() #'! l-!9% #0'1 4, l-!9% #1 "L*4 A%0 activity lo
pyExcel C+l 2' LB+%
py
py4.13 #2" 2# #1 "L*4 A%0 2# 1 %8H!
py
py+ l2 Asset Registry C l loadAssetRegistry() B+% l-!9%-8 # LA%0 %
pyrenderRegistryContent() A* #2" 2#A 1 %8H! 9lC l 5H!5*4 4L*2!2
py@ %5H" 7H- %8H!, % %8H! A%0"l2"-8 # L@ l2 %8H! H2 assignDev
pyopenAddGroupModal(), openRenameGroupModal(), confirmGroupActio
py
py4.14 2# 1 2# 9lC lA%0 Activity Log
py
pyAdmin *2!2# @ 4 + l2 1 2# 9lC l@ 7H-@ %5H" Role, l-!9% profile /
py#0 !5 2##l2 activity log @!7H-@ 4 @+ 8 2# L*3 1 @ H Login, 1
pyA ID 9lC l, % l-!9% A%0 export/import
py
py4.15 Report
py
pyprintReport() B+% #0'1 4 2# 3#8 - D LA%l'C+l 9lC l@%7- #2" 2# 5
pygenerateSelectedReport() *#l2 HTML #2" 2 B "#'! l-!9%-8 # L, asse
py#9 H- /+%1 H-!, 9l#1 #2 A%0 9lA l H- *H C+l buildReportDocum
py@ 7H-*#l2 @- *2#*3+#1 4! L/ 1 6
py
py5. Flow 2# 3 2 +%1

```

py
py5.1 Flow Login
py
py'text
py 9IC I Login
py-> Firebase Google Auth
py-> onAuthStateChanged 3 2
py-> -H2 users/{email}
py-> 3+ role/profile/allowedSites
py-> applyRoleRestrictions()
py-> switchSite(default)
py-> setupRealtimeListener(site)
py-> A* appContent
py'
py
py5.2 Flow 1 6 @+ 8 3#8
py
py'text
py %4 -8 # L A 1 / 2#2
py-> openForm(device)
py-> B+% assetInfo + records
py-> 9IC I #- * 2 0/#2"%0@-5" /D %L
py-> saveData()
py-> #' *4 4LA%0 validate
py-> uploadFileToStorage() 12!5D %L
py-> saveDeviceRecords()
py-> createLog()
py-> updateDeviceSummary()
py-> updateDeviceStatusOverlays()
py-> sendEmailNotify() 2!@ 7H- D
py'
py
py5.3 Flow Dashboard/Overlay
py
py'text
pyFirestore records @ %5H"
py-> onSnapshot()
py-> updateDeviceSummary()
py-> renderDashboardCharts()
py-> updateDeviceStatusOverlays()
py-> 9IC I@+G 2#2 / #2 /A 1 %H2*8
py'
py
py5.4 Flow Import Excel
py

```

```

py'text
py 9IC I@%7- D %L .xlsx
py-> FileReader -H2 D %L
py-> XLSX A % sheet @ G array/json
py-> map -%1! L@ G assetInfo/records
py-> processAndSaveImport()
py-> batch commit D Firestore
py-> refresh 2#2 A%0 overlay
py'
py
py6. '2!+!2" - I-!9%*3 1 C Record
py
pyRecord C records C I@ G @+ 8 2# L - -8 # L B " field 5H D I @ H
py
py Field '2!+!2"
py --- ---
py customId #+1* @+ 8 2# L-1 B !1 4 @ H KPL-000001
py status * 2 0@+ 8 2# L @ H down, abnormal, ok
py brokenDate '1 5H /A I @+ 8 3#8
py description #2"%0@-5" -2 2#@*5"
py brokenFileUrl URL D %L/#9 H- H-!C Storage
py fixedDate '1 5H H-!@*#G
py solution '4 5A ID / % 2# H-!
py fixedFileUrl URL D %L/#9 +%1 H-!
py repairCost H2C I H2" H-!
py acknowledgedBy 9I#1 #2 @+ 8 2# L
py acknowledgedAt @'%2#1 #2
py
py7. 8 @ H - B #@
py
py- C I@'G + I2@ 5"' 3C+I deploy H2"
py- @ 7H-! Firebase B " # D!H I- !5 Backend A"
py- #- #1 *4 4L 9IC I+%2"#0 1
py- A* * 2 0-8 # L A 1 #4 H2 image map A%0 overlay
py- !5 I-!9% #1 "L*4 #I-! #0 1
py- #- #1 Import/Export Excel A%0#2" 2
py- !5 Realtime update 2 Firestore
py
py8. I- '##0'1 / I-@* -A 0
py
py1. Firebase config -"9HC Client @ G @#7H- 4 - Firebase Web
pyA H I- ' 8! '2! %- 1" I'" Firestore/Storage Security Rules C+I#1 8!
py2. I-!9% records 9 @ G @ G array C @- *2#@ 5"' +2 record !2 6I
py-2 I- 3 1 2 @- *2# Firestore '# 4 2# 2A" @ G subcollection C -
py3. Logic 3 ' !2 -"9HC main.js D %L@ 5"' 3C+I 9A%"2 '#A" modul

```

```

firestore, assets, reports, registry, charts
4.  '# "I2" H2 config/#2" 7H-D L/#2" 2#-8 # L D @ G D %L config +#
@ 7H-A ID D I H2"
5.  '#@ 4H! test + #7- validation layer @ 4H!@ 4! B "@ 20 import Ex
A%0*4 4L 2# 1 6
9.  *#8
B #@ 5I 7-#0 1 2# I-!9%-8 # L Microgrid/BESS A Frontend-only
@ G Backend-as-a-Service 8 @ 7H-!B" +%1 7- index.html
3+ I2 5H@ G + I2 -A%0B+% D% #2#5 *H' main.js ' 8! state, *4 4L
Firestore/Storage, 2#*#8 I-!9%, 2#A* A 1, 2# import/export A%
8 5@ -#L@ 7H-! H2 1'A #D L 1 8 1 (currentSiteKey) A%0@- *2#-
sites/{siteKey}/devices/{deviceName}

```